

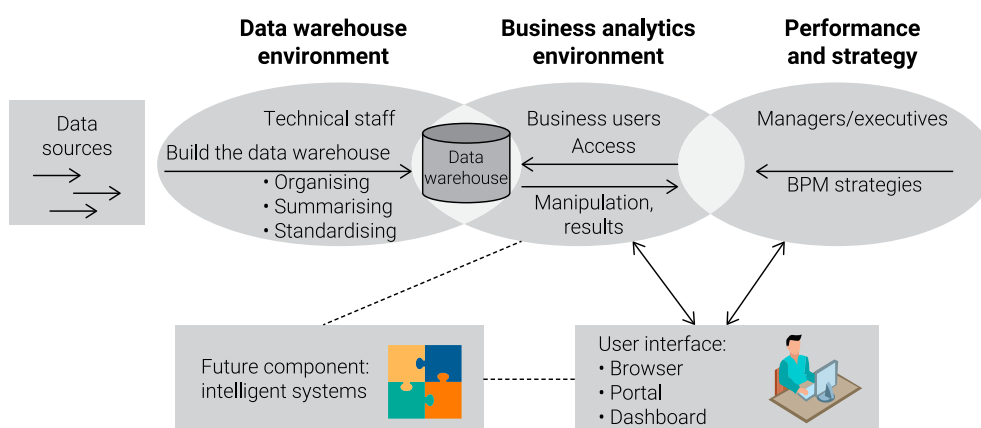
In this topic, you will look at business analytics (BA) architecture and explore the components of this architecture to understand how business analytics can be applied within an organisation. Understanding these components will enable you to make the right decisions when you undertake a business analytics project for your company.

BA architecture

The image below provides a high-level view of a BA architecture. There are various environments or components to the architecture such as the data warehouse environment, the business analytics environment and the performance and strategy derived from the two environments. The data warehouse environment is where the data is stored, organised, summarised and standardised. This data is then analysed in the business analytics environment to generate insights. The insights generated then help in building the business performance management (BPM) strategies. The user interface for the business analytics environment and for performance and strategies are systems that the analysts and other users will use to perform functions. The interface comprises browsers, portals, dashboards and so on. There is also an opportunity for business users to build intelligent systems and integrate them in the business architecture for future use.

Figure 3.1

BA architecture



(Click on the image to expand)

© Sharda et al., 2014

[\[Image transcript\]](#)

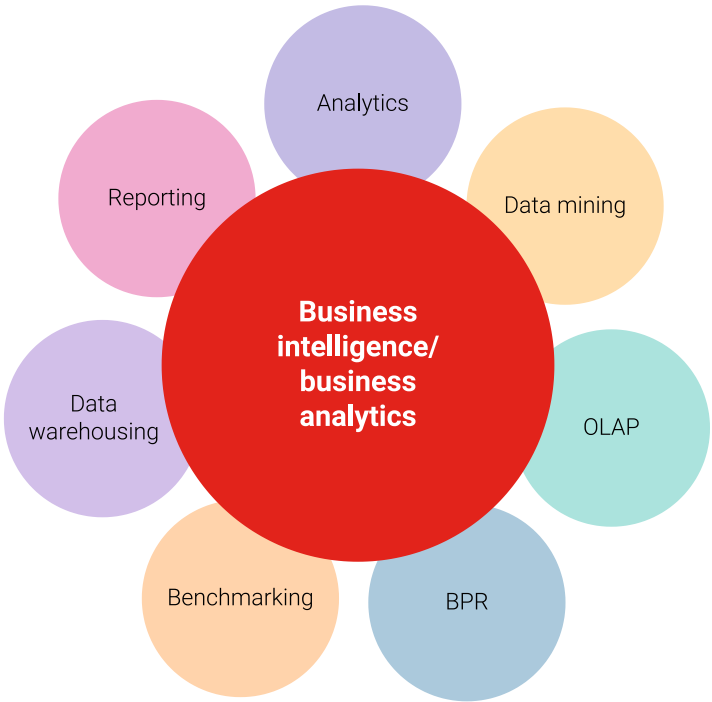
Business intelligence / business analytics is an umbrella term that includes architectures, tools, databases, applications and methodologies. These can be grouped into:

- data warehouses and data marts
- online analytical processing (OLAP)
- querying and reporting
- analytics and data mining
- business performance management
- dashboards and benchmarking.

It can be represented as shown in the image below.

Figure 3.2

Groups of business intelligence and business analytics



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Let’s explore each of these components in detail.

Data warehouse

The key terms that you will often hear being used in business analytics are data warehouse, data mart and data lakes. While they all relate to data, they have different meanings and applications in business analytics. In this topic, you will learn more about data warehouse.

A data warehouse is the cornerstone of medium-to-large business intelligence / business analytics systems. It eradicates organisational silos to present a single version of the truth. In the past, it included only historical data that was organised and summarised for producing end-user reporting. Today, it includes access to current data for real-time decision support. A data warehouse is structured on the dimensional model of a business process and is also the source for querying, reporting and OLAP.

Data mart

Data mart is a pattern used in the data warehouse environment to retrieve data about a particular client.

Data lake

A data lake is a centralised repository that allows you to store all your structured and unstructured data at any scale. You can store your data as is, without having to first structure the data, and run different types of analytics — from dashboards and visualisations to big data processing, real-time analytics and machine learning to guide better decisions.

The main differences between a data warehouse and a data mart are shown in the image below.

Table 3.1
Data warehouse vs data mart

Data warehouse	Data mart
Data warehouse stores the data from multiple subject areas.	Data mart holds the data related to a particular area such as finance, HR, sales, etc.
It is a central repository of data in an organisation.	It is the subset of a data warehouse.
Data is integrated into a data warehouse as one repository from various sources.	Data is integrated into a data mart from fewer sources than a data warehouse.
A data warehouse is usually modeled from a fact constellation schema.	Data mart is designed focused on a dimensional model using a star schema.

Data warehouse	Data mart
It is difficult to design and use a data warehouse for its size, which can be greater than 100 Gigabytes.	It is comparatively easier to design and use data mart because of the flexibility of its small size.
Data warehouse is designed for decision-making in an organisation.	Data mart is designed for specific user groups or departments.
It follows a top-down approach.	It follows a bottom-up approach.
Data warehouse holds less de-normalised data than a data mart.	Data mart stores highly de-normalised data.

© Pedamkar, 2020

The major differences between a data warehouse and a data lake are mentioned in the image below.

Table 3.2*Data warehouse vs data lake*

Characteristics	Data warehouse	Data lake
Data	Relational from transactional systems, operational databases, and line of business applications	Non-relational and relational from IoT devices, web sites, mobile apps, social media, and corporate applications.
Schema	Designed prior to the DW implementation (schema-on-write)	Written at the time of analysis (schema-on-read)
Price/performance	Fastest query results using higher cost storage	Query results getting faster using low-cost storage
Data quality	Highly curated data that serves as the central version of the truth	Any data that may or may not be curated (i.e. raw data)
Users	Business analysts	Data scientists, data developers, and business analysts (using curated data)
Analytics	Batch reporting, BI and visualisations	Machine learning, predictive analytics, data discovery and profiling

© Amazon Web Services, Inc. 2019

Data mining

Data mining is an umbrella term used to refer to a suite of computer algorithms from fields such as statistics, machine learning, data mining and artificial intelligence. Data mining transforms data into insights. It identifies causes, makes predictions and enables decision optimisation. Data mining is an emerging field of big data analytics where unstructured data, such as text, images and audio/video, are also introduced into the decision-making process and it overcomes the silos created by the nature of data.

Business performance management

Business performance management (BPM) is also referred to as corporate PM or enterprise PM. It is an umbrella term used to describe the methodologies, metrics, processes and systems used to monitor and manage the business performance of an enterprise in achieving one or more pre-selected goals. It encompasses all the processes, information and systems used by managers to set

strategy, develop plans, monitor execution, forecast performance and report results with a view to achieving sustainable success no matter how success may be defined. The activities within BPM run on a loop as shown below:

Measurement → Evaluation → Adjustment

Some of the methods used in BPM are Six Sigma, balanced scorecard and activity-based costing.



Read

Accenture is one of the world's leading consulting companies that help organisations grow. This is an article by Accenture on the importance of business performance analytics. Read this article to know more about how business analytics can improve an organisation's performance and bring in revenue.

[Business performance analytics: What's old, what's new, what's tried-and-true?](#) (Accenture, 2019)

Dashboard and benchmarking

A dashboard provides a comprehensive graphical/pictorial view of corporate performance measures, trends and exceptions. This can extend from organisational level to department and all the way down to employee. Dashboards facilitate benchmarking across the hierarchy. A dashboard will retrieve data in real time from a data warehouse or reporting interface. Some categories of dashboards are:

- strategic, analytical, operational
- quantitative, qualitative.

Evaluating analytics tools — Gartner Magic Quadrant

Gartner Magic Quadrant is used to evaluate the various analytics tools. It provides a graphical competitive positioning of four types of technology providers in fast-growing markets, measured across completeness of vision vs ability to execute. The four types are:

Click on the below headings to reveal further information.

› Leaders

› Challengers

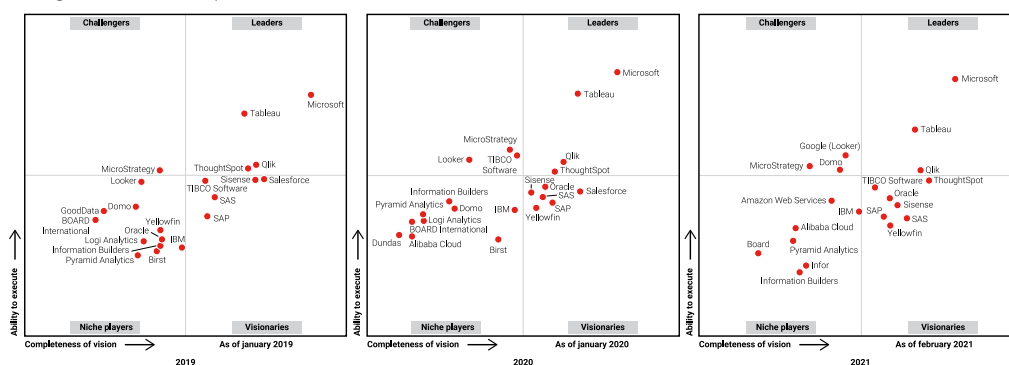
› Visionaries

› Niche players

Gartner Magic Quadrant is updated annually and the latest update was in February 2021. Below are examples of the latest models published for analytics and BI, and data science and machine learning.

Figure 3.3

Gartner Magic Quadrant: Analytics and BI



(Click on the image to expand)

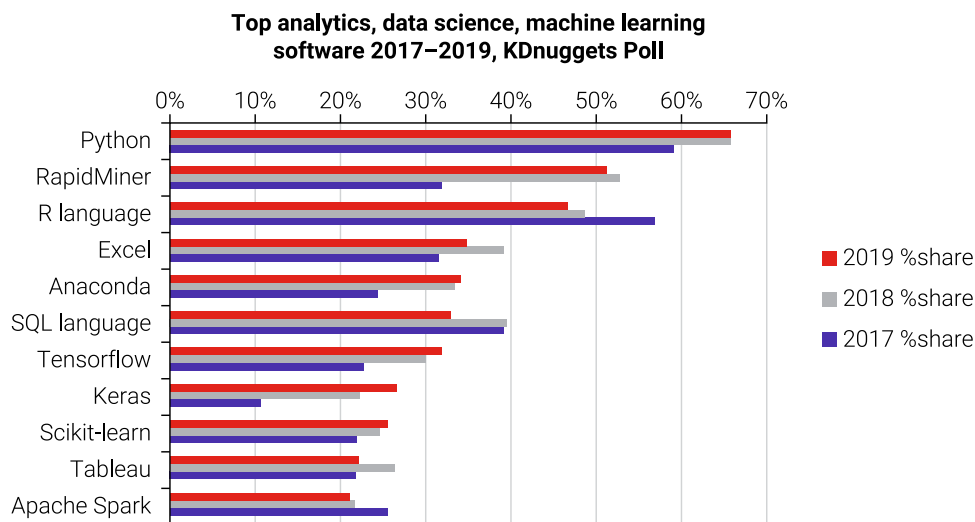
© Gartner, 2021

Figure 3.4*Gartner Magic Quadrant: Data science and machine learning*

(Click on the image to expand)

© Gartner, 2021

As a data analyst, you must be aware of the different analytics tools currently available. The image below shows the number of tools and their market share, based on their usage and adoption.

Figure 3.5*Number of data tools and their market share*

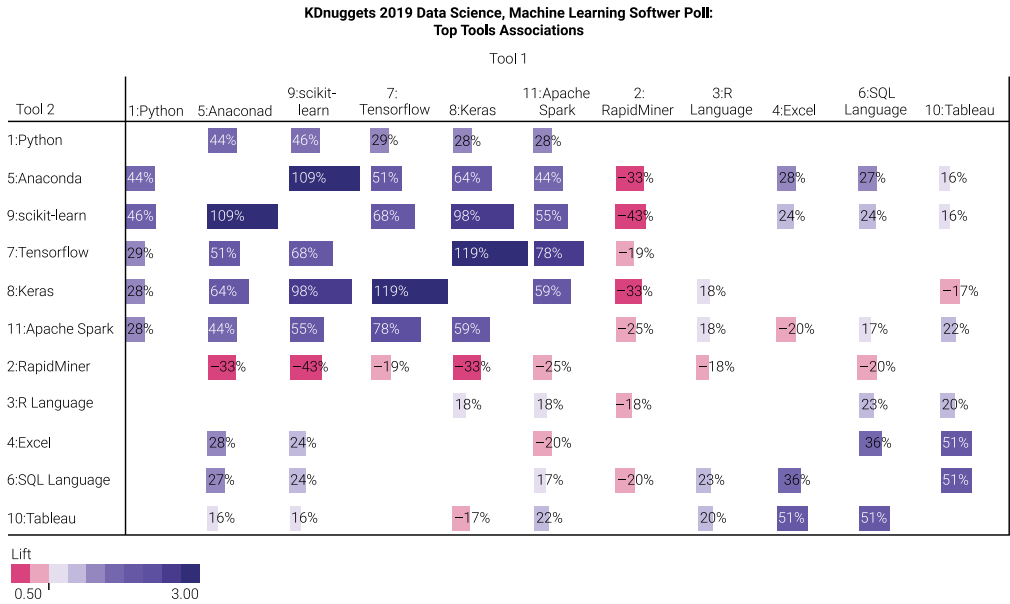
(Click on the image to expand)

© Piatetsky, 2019

[\[Image transcript\]](#)

The associations between the different tools are often used in the data science and machine learning space. These associations are shown in the image below that highlights positive association (green) or negative association (red). The higher the percentage, the stronger the degree of association.

Figure 3.6*Associations between different data tools*



(Click on the image to expand)

© Piatetsky, 2019

[\[Image transcript\]](#)

However, it is important to note that the association does not necessarily mean that all tools were used together on each project, but having knowledge and skills to use both tools X and Y makes it more likely that both X and Y were used together on some projects.

Another important statistic to know about is the types of data that are being analysed. The image above shows an increment in unstructured data (e.g. anonymised data, location/geo data, images/video data) being analysed. Some of the most popular data types analysed in 2017 were:

- tabular data (fixed number of columns), 69.8% — first place, as in the past polls
- textual data, 46.4% — moved to 2nd place compared to 2014
- time series data, 45.6% — dropped to 3rd place
- JSON, 25.5% — up from 7th place
- anonymised data, 22.8% — up from 10th place
- location/spatial data, 22.6%.

 **Watch**

Top 10 data science tools over time

This 2-minute video [1.49 min.] to shows the top 10 data science tools from 2000 to 2019 in terms of tool usage rates by data scientists on KDnuggets.com. As a data analytics professional, it is good to be aware of the popular tools and their functionalities.

Top 10 Data Science Tools Over Time



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